

SinoBF-1: Satellite mapping of every building's function in urban China reveals deep built environment disparities

User guide V1.0 (March. 2th, 2026)

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Or check our data website: <https://doi.org/10.5281/zenodo.17844789>

Note: This document only provides a brief introduction to the SinoBF-1 data, emphasizing the organization, download, and use guide of the data. We strongly suggest checking **our paper** which includes more detailed information. Paper: <https://www.nature.com/articles/s41467-026-69589-5>

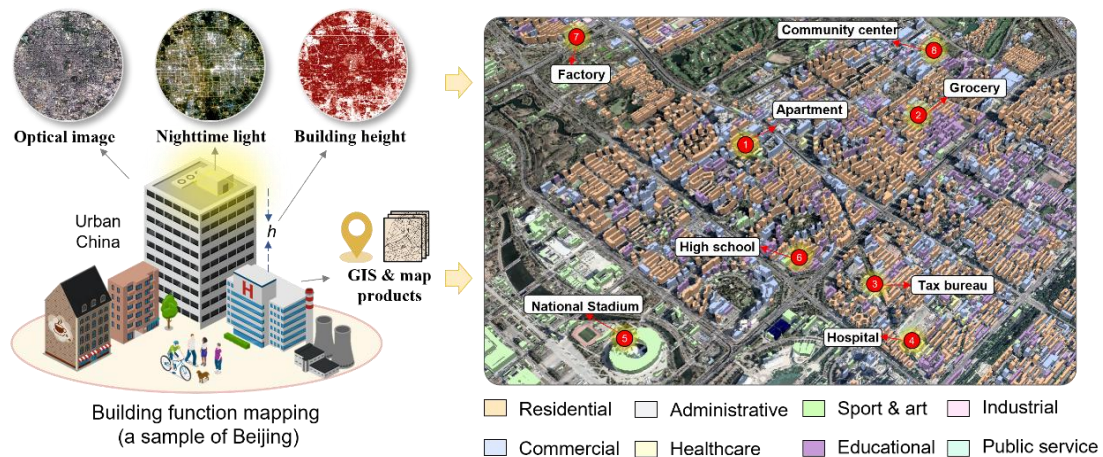


Figure 1. Demonstration of the sample areas of Beijing.

Introduction:

We present SinoBF-1, a national-scale building functional map of China derived from more than 69 terabytes of multi-source satellite data, including optical imagery, nighttime lights, and altimetry observations. This fine-grained dataset captures the attributes and functions of approximately 110 million buildings across 109 cities, covering a total area of 1,094,344 km².

We utilized three modalities of data: high-resolution optical images (1.07 m/pixel) from Google Earth for fine-scale building appearances, 10-meter building height data from China's CNBH-10m dataset, and 10-meter nighttime light imagery from the SGDSAT-1 satellite to assess nighttime activities and energy consumption patterns. To ensure stable labels for model training, we used the CN-OpenData dataset and the East Asia Building Dataset to define the boundaries of building footprints, whereas land-use and area-of-interest (AOI) data from OpenStreetMap were integrated to generate labeled samples for building function. Finally, we employed the Paraformer model, a CNN-Transformer model we proposed in CVPR 2024 <https://github.com/LiZhuoHong/Paraformer> to identify every building's function in urban China. We conducted validations using two independent sources: (1) official reports from the National Bureau of Statistics and the Ministry of Housing and Urban-Rural Development of China; (2) large geospatial reference point sets comprising 5,280,695 samples from Amap Inc. The functional map achieved strong agreement with external benchmarks, with an average R^2 of 0.8208 at the statistical level and an overall accuracy of 84.65% at the individual building level (see paper for details). A case of application with the SinoBF-1 is shown in Figure 4.

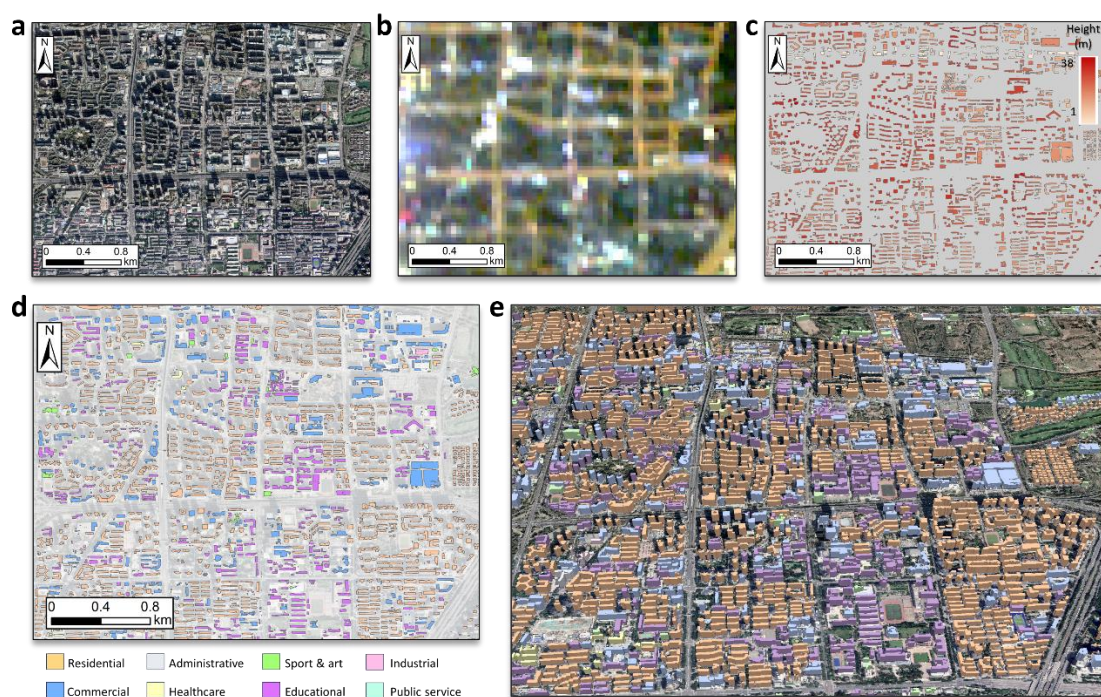


Figure 2. A samples of the multi-modal input data source (a–c), the produced 2D building function map (d), and the 3D building function map (e).

Data organization:

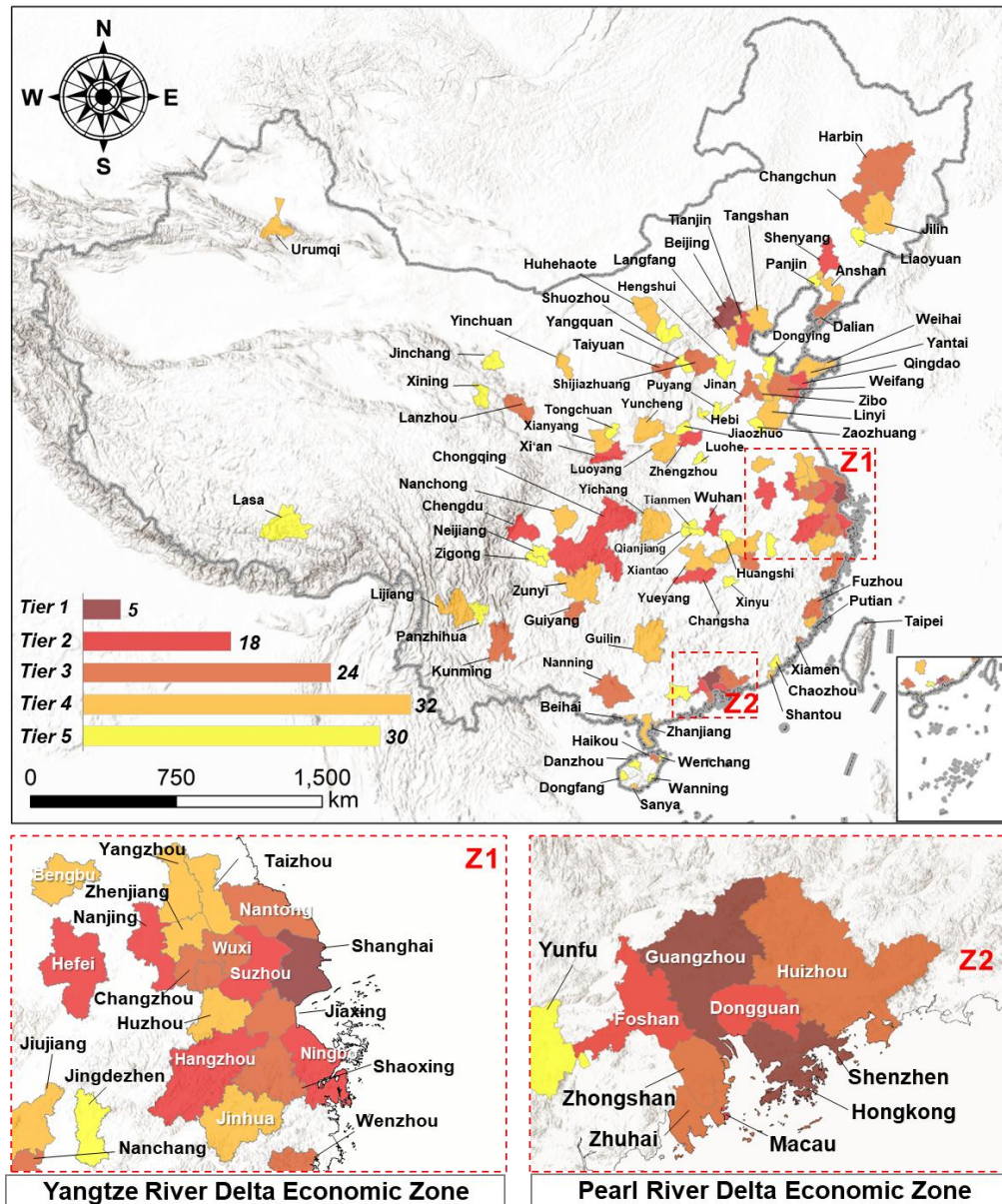


Figure 3. Administrative boundaries of the 109 major cities in China

This version of the dataset includes building-level functional maps of 109 Chinese cities, organized in the ESRI Shapefile format. The .shp file for each city is stored in a “.zip” archive. Each city file is named **G_P_C.zip**, where: **G** indicates the geographical region (South, Central, East, North, Northeast, Northwest, Southwest); **P** indicates the provincial administrative region; and **C** indicates the city name.

For example, the building functional map for Wuhan City, Hubei Province, is named Central_Hubei_Wuhan.zip. Furthermore, each shapefile contains building functional types stored in the attribute "Function", with the corresponding building functions listed below (Table 1):

Table 1 The function definition, and typical subclass of each building function type.

Legend	Function	Definition	Typical subclass
1	Residential	Residential buildings refer to a structure and its supporting facilities designed for people's daily living. This category includes low-rise residential buildings, complete apartment complexes, semi-detached villas, and self-built houses for original rural residents.	Residential district Residential Villa district Domestic premises Dormitory Accommodation
2	Commercial	Commercial buildings are designed for activities related to commerce and service industries. It accommodates various business operations such as retail, offices, hotels, and entertainment venues. This category includes commercial complexes, office buildings, service industry facilities, and tourism-related buildings.	Convenience store Shopping mall Supermarket Company Coffee shop Hotel Market Food Beverage
3	Administrative	Administrative buildings serve as the office space for government agencies, political organizations, and related institutions. This includes offices of the National People's Congress, the Chinese People's Political Consultative Conference, courts, procuratorates, public security bureaus, fire departments, customs, and other governmental bodies.	Court Government office building Bureau of public security Customs Procuratorate
4	Healthcare	Healthcare buildings are designated for medical and public health services. It includes hospitals, outpatient departments, clinics, centers for disease prevention and control, health supervision centers, chronic disease treatment institutions, emergency centers, blood donation centers, and family planning service centers.	Specialized hospital Pharmaceutical sales First aid centre General hospital Disease prevention Clinic
5	Sport and art	Sport and art buildings encompass cultural and recreational facilities such as libraries, exhibition centers, museums, radio and television studios, cinemas, art groups, music halls, and various cultural activity centers. They also include sports facilities such as stadiums, arenas, and training bases.	Fitness center Cultural palace Museum Memorial hall Sports center Stadium
6	Educational	Education buildings include facilities for academic, research, and training purposes. This category covers universities, colleges, secondary and vocational schools, special education schools, as well as institutions for scientific research, surveying, and testing.	University School Kindergarten College Laboratory Institute
7	Public service	Public service buildings provide essential community and infrastructure services. These include postal facilities, utility stations, telecommunications centers, sanitation services, water conservancy buildings, and other municipal support facilities.	Post office Community center Service station Gas station Sanitation station Beadhouse Power supply station
8	Industrial	Industrial buildings are used for production, research, and logistics related to industry. It includes factories, research plants, headquarters, manufacturing areas, and warehouses for storage and distribution.	Warehouse Industrial manufacturing Logistics and warehouse Industrial building Factory

Note: Due to Zenodo's limit of 100 files, the cities in Hubei Province and Hainan Province were grouped into single compressed packages (i.e., Central_Hubei.zip and South_Hainan.zip).

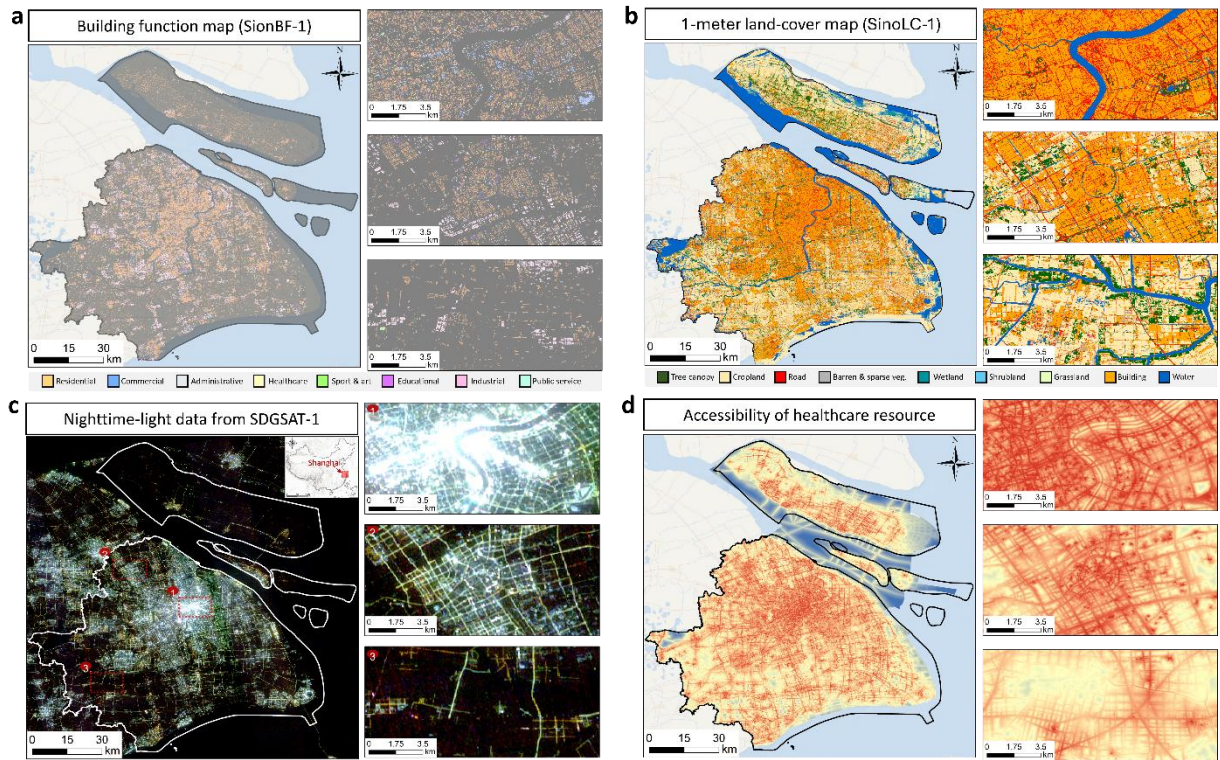


Figure 4. The 1-meter resolution accessibility maps of Shanghai, produced by combining the building function map, the traffic network data, and the 1-meter resolution land-cover map (SinoLC-1). (a) The building function map produced in this study. (b) The SinoLC-1 map produced in our previous study. (c) The 10-meter resolution nighttime light data collected from SDGSAT-1 satellite, indicating the main activity areas of residents. (d), The accessibility maps of healthcare resources in Shanghai.